Robotics Assembly Line - Create3

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Assembly Line

Robotics Assembly Line is a highly automated production system where robots performs various tasks. In this case, a dice is used to simulate what happens in a real-world automated factory.

Key Components of a Robotics Assembly Line

1. **Robots** - Collaborative Robots are used as they can work alongside humans in any environment safely (Fanuc CRX10, Standard Robots, Universal Robots).

2. **Human Machine Interface (HMI)** - This is important for monitoring, controlling, and visualizing industrial processes. **Ignition** is used to collect real-time data from sensors, robots, PLCs, and other devices on the assembly line. It also provides visual dashboards and screens for operators to view the status of the assembly line. It also allows the operators to send commands to the robots and other devices, like Start, Stop, **E-STOP**. And Ignition has various database connections like PostgreSQL, MySQL, and many more to store historical data, and can be used for analysis and to generate reports.
   Some Key Features of Ignition:

   - Monitoring - Ignition SCADA is used in this project for real-time data collection, visualization and control.
   - Command - This feature allows operators to be able to snd commands like Start, Stop, E-STOP, and also adjust settings.
   - Database Integration - Ignition allows for connection to databases like PostgreSQL, MySQL, Microsoft SQL Server for historical data storage and analysis.
   - Remote Access - Ignition provides various ways to access Ignirion Vision and perspective projects from a remote location by using the Gateway's IP Address, or using VPN, or Port Forwarding, or Reverse Proxy.

3. **End-of-Arm Tools (EOAT)** - Tools used for specific tasks like gripping, screwing, welding, etc. The OnRobot Gripper is used.

4. **Sensors and Vision Systems** - This systems are used to ensure precision by detecting part positions, orientations, and quality, and they help to guide the robots for accurate assembly. The OnRobot Eyes, Standard Robot built-in camera, external camera for the Fanuc, RoboCop Industrial IoT Device, Industrial IoT Devices and Lights. All these are used to ensure accuracy in the assembly line.

5. **Control Systems** - Programmable Logic Controllers (PLCs) and computers are used to manage the workflow. The Industrial IoT Devices and Lights is used to control the system.

6. **Conveyor Systems** - These are used to transport parts and products between workstations to ensure seamless operations. Two conveyors are used for this process.

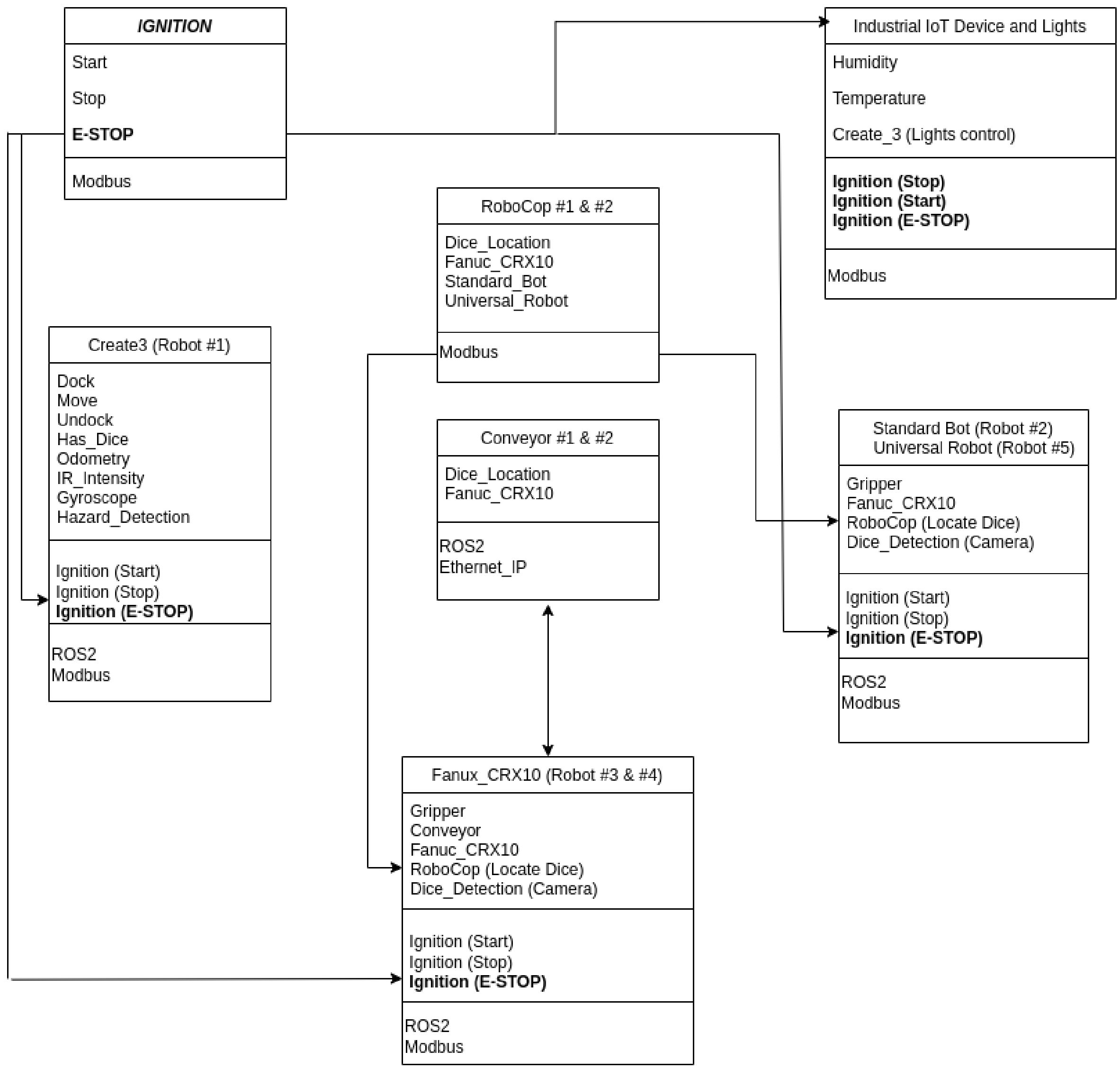

Figure 1: Assembly Line Class (UML) Diagram

Create3

Create3 is a robot that has a full suite of on-board sensors and actuators. The robot provides autonomous behaviors out of the box, like docking, wall-fllow and reactions to obstacles.

Key Components of Create3

1. Integrated Sensors - Create3 has various sensors like Cliff sensors, wheel drop sensors, and bump sensors, all necessary for obstacle detection and avoidance.
2. Navigation and Localization - Create3 supports Simultaneous Localzation and Mapping (SLAM), this enables the create to gather information about it's environment by building a map of it's environment and localize itself within the map. It also has an in-built odometer used for precise movement tracking.

Modbus Integration

To successfully integrate Modbus into the Assembly Line system, ROS2 was used to read/write data to various Modbus registers, and to ensure efficient and real-time communication requires a multi-threading approach.

This was used to allow the ROS2 nodes to handle asynchronous tasks (like the reading and writing operations to the various Modbus registers) without blocking the main execution thread.

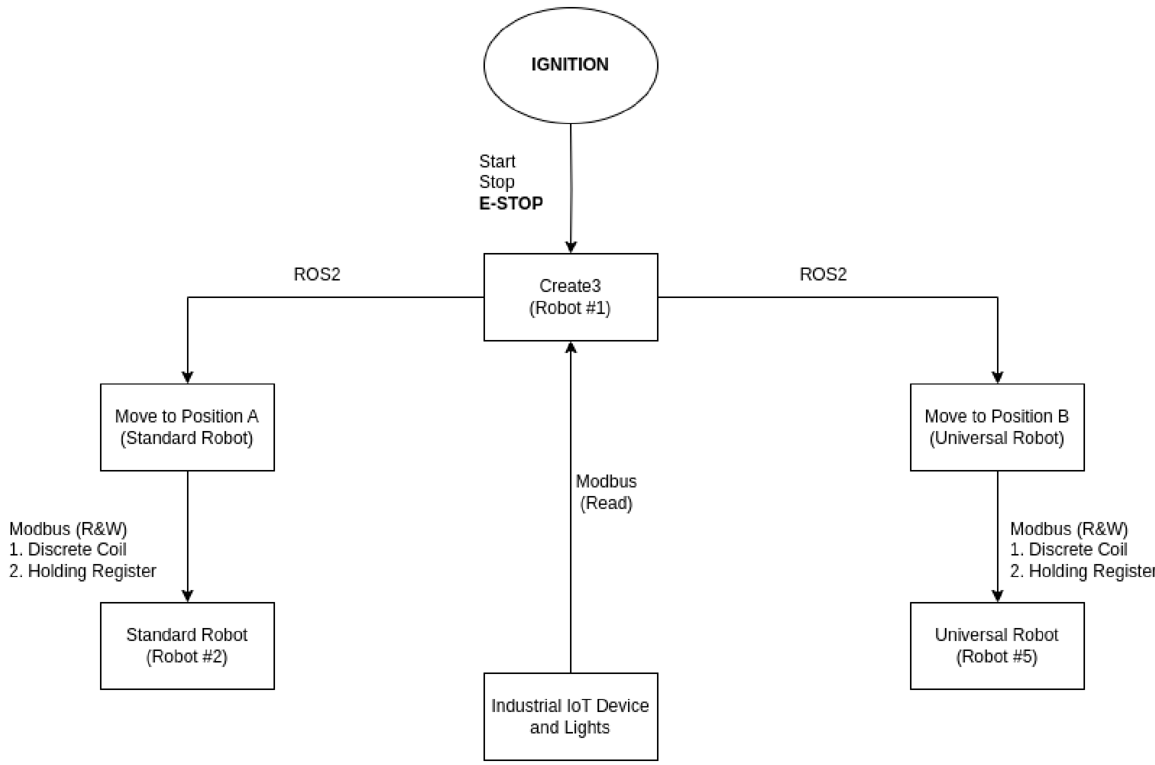

Figure 2: Create 3 Block Diagram